Project Proposal

Automated Accessibility Audits: Improving Software Quality for Everyone

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Introduction

In the study of Software Quality Assurance, we often focus on functional correctness, performance, and security. However, a critical yet frequently overlooked quality attribute is Accessibility . Accessibility ensures that digital products are usable by everyone, including individuals with visual, auditory, motor, or cognitive impairments.

As the web becomes more central to our daily lives, "Quality" must include "Inclusivity." This project focuses on the practical implementation of automated accessibility audits. My objective is to demonstrate how modern SQA tools can be used to identify barriers to inclusivity and ensure that web applications meet international standards like the Web Content Accessibility Guidelines (WCAG 2.1).

Problem Statement

Today, most websites are designed only for average users. This leaves out many people who use special tools to browse the internet, such as screen readers or just a keyboard.

Checking every page for these problems by hand is very slow. It is also easy to make mistakes, especially on large websites. Without automatic checks in place, these accessibility problems often go unnoticed until the website is finished. This makes the software impossible to use for many people.

There is a clear need for a fast, automatic way to find and fix these quality bugs early, while the software is still being built.

Project Goals

The main goal of this project is to show a clear process for testing website accessibility automatically.

My specific goals are:

- To get a starting score for the website to see how accessible it is at the beginning.
- To find accessibility errors and group them by how serious they are (from Critical to Minor).
- To fix the problems in the code and then test the site again to prove it has improved.
- To show that accessibility is something we can actually measure and make better, just like any other part of software quality.

QA Tools & Technologies

Playwright (with axe-playwright): I will use Playwright to automatically open the website and move through different pages. I will also use a tool called **Axe-core** to scan the website's code and find hidden accessibility problems.

Google Lighthouse: I will use this to give the website a score from 0 to 100. This score will show me exactly how much the website's quality improves after I fix the errors.

pa11y: This is a fast tool that I will use to create clear reports. These reports will list every accessibility issue found on the site.

Target Application: My goal is to build an application that clearly shows how these tools can identify and help resolve various quality issues.

Conclusion

This project shows how we can stop just reacting to problems and start preventing them from the beginning. By focusing on accessibility, I am looking at a very important part of software quality that is often forgotten. I believe this project will show that with the right tools, we can build software that works well and is easy for everyone to use.